

NEW BRAUNFELS, TX, USA

WMO: 722416

Lat: 29.709N

Lon: 98.046W

Elev: 197

StdP: 98.99

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 12971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.1	-11.0	1.5	5.1	-8.7	1.8	7.6	11.9	12.4	11.0	12.0	3.7	20	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	37.9	23.4	37.0	23.5	36.0	23.5	25.7	31.9	25.3	31.4	25.1	31.0	4.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.8	27.0	24.0	19.4	26.8	23.7	19.0	26.7	80.1	32.2	78.5	31.5	77.0	30.9	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.6	9.2	8.3	DB	-5.3	39.9	2.2	1.9	-7.0	41.2	-8.3	42.3	-9.5	43.4	-11.1	44.7	
			WB	-6.8	26.4	2.0	0.5	-8.2	26.8	-9.4	27.1	-10.5	27.4	-11.9	27.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	11.0	13.4	17.1	21.1	24.7	27.9	28.8	29.6	26.7	21.7	16.2	11.8
	DBStd	7.63	5.08	5.60	4.77	3.82	3.10	1.91	1.70	1.71	2.82	4.29	5.05	4.99
	HDD10.0	129	48	28	6	1	0	0	0	0	0	1	9	38
	HDD18.3	813	231	154	80	17	2	0	0	0	1	19	99	210
	CDD10.0	4094	80	122	227	332	454	538	584	607	499	362	194	94
	CDD18.3	1736	5	15	43	99	198	288	326	349	250	123	33	8
	CDH23.3	18657	39	109	302	810	1844	3240	3839	4402	2628	1144	251	48
	CDH26.7	9159	3	21	68	255	764	1646	2094	2524	1313	421	47	4
Wind	WSAvg	3.9	4.1	4.4	4.5	4.6	4.3	3.7	3.4	3.4	3.2	3.5	3.7	3.9
Precipitation	PrecAvg	888	50	57	50	67	119	108	46	68	91	108	74	62
	PrecMax	1484	247	231	133	231	401	357	148	266	267	680	274	445
	PrecMin	471	0	0	3	2	2	0	0	0	0	8	0	0
	PrecStd	275	46	43	28	49	92	90	43	61	60	109	62	77
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	29.5	30.7	33.8	35.8	38.5	38.8	39.3	38.3	34.0	30.1	27.1
		MCWB	16.6	17.7	19.0	20.2	23.4	23.2	23.1	23.6	22.6	21.8	21.4	18.7
	2%	DB	24.2	26.5	28.6	31.3	34.0	36.5	37.4	38.0	36.3	32.3	28.1	24.5
		MCWB	16.4	17.9	18.5	20.9	23.0	23.4	23.5	23.6	23.1	22.3	19.7	17.5
	5%	DB	22.2	24.2	26.7	29.4	32.4	35.1	36.3	37.2	34.7	30.9	26.4	22.7
		MCWB	15.7	16.6	18.7	20.4	23.1	23.6	23.7	23.6	23.0	21.5	18.9	17.2
	10%	DB	20.1	22.2	24.7	27.8	31.0	33.7	35.1	36.2	33.0	29.1	24.4	20.8
		MCWB	15.2	16.6	17.8	20.0	22.7	23.7	23.8	23.7	22.8	20.8	18.6	16.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	20.9	22.0	24.1	25.7	26.3	26.0	25.8	25.8	25.2	23.3	21.2
		MCDB	22.7	24.5	26.1	29.5	31.7	32.7	32.8	33.0	30.9	29.0	26.8	24.1
	2%	WB	19.2	20.0	21.0	23.0	25.0	25.7	25.5	25.5	25.2	24.4	22.1	20.2
		MCDB	21.9	23.3	24.9	27.8	30.6	31.7	32.0	32.2	30.3	28.2	25.3	22.8
	5%	WB	17.9	19.2	20.3	22.3	24.3	25.2	25.1	25.1	24.8	23.8	21.1	19.1
		MCDB	20.7	22.2	24.0	26.7	29.5	31.0	31.4	31.7	29.9	27.6	24.4	21.6
	10%	WB	16.1	18.1	19.6	21.7	23.8	24.8	24.8	24.8	24.3	23.0	20.1	17.5
		MCDB	19.0	20.9	23.1	25.7	28.6	30.3	30.7	31.2	29.4	26.9	23.5	19.8
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.3	12.4	12.1	11.6	11.5	12.2	12.9	12.2	12.7	12.5	12.4
		MCDBR	15.4	14.6	13.6	13.6	12.9	13.3	14.1	14.4	13.9	13.1	13.4	14.8
		MCWBR	9.1	8.3	6.7	5.9	4.5	3.3	3.1	3.1	4.0	5.3	7.0	9.3
	5% WB	MCDBR	11.6	10.9	10.8	10.1	10.4	10.6	11.4	12.4	11.1	10.3	10.2	11.7
	MCWBR	8.2	7.5	6.4	5.1	4.3	3.4	3.1	3.0	3.6	5.0	6.1	9.0	
Clear-Sky Solar Irradiance	taub	0.335	0.335	0.362	0.401	0.449	0.430	0.441	0.427	0.416	0.375	0.363	0.319	
	taud	2.471	2.482	2.418	2.325	2.238	2.343	2.344	2.368	2.351	2.452	2.420	2.533	
	Ebn at Noon	899	933	925	894	848	860	848	860	858	877	858	902	
	Edh at Noon	93	100	114	129	141	127	126	122	120	102	97	83	
All-Sky Solar Radiation	RadAvg	3.01	3.50	4.47	5.34	5.78	6.58	6.62	6.30	5.19	4.31	3.24	2.67	
	RadStd	0.33	0.63	0.43	0.51	0.53	0.42	0.56	0.35	0.50	0.57	0.36	0.34	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+146
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86

Nomenclature: See separate page